

Single-cell splicing analysis with proximal NND models

1 Data objects

The prompt-provided Zenodo object, `splice_adata_for_figures_mouse_foundation.h5ad`, is a fitted mouse splicing atlas. It contains cell metadata, learned cell program scores `obsm["X_PHI"]`, and learned junction loadings `varm["psi_learned"]`, but it does not contain the raw count layers needed to construct per-cell PSI estimates. In this object, the learned cell program matrix has shape 142315×20 , and the learned junction loading matrix has shape 70102×20 . Equivalently, the cell metadata contains columns `SP_1, ..., SP_20`. Thus the fitted mouse atlas has 20 splicing programs.

The raw count object used for model fitting was

`/media/david/HDD/splicing_data/model_ready_aligned_splicing_data_20251009_024406.h5ad`.

This object has shape 142315×89831 . Its `X` matrix, `layers["cell_by_junction_matrix"]`, and `layers["cell_by_cluster_matrix"]` are all stored as HDF5-backed CSR matrices. Because the count layers contain about 1.28×10^9 nonzero entries, the analysis avoids loading the full AnnData object with `ad.read_h5ad`. Instead, selected row and column blocks are read directly from the CSR arrays `data`, `indices`, and `indptr` using `h5py`.

2 Block selection

The analysis first selects a manageable dense modeling block from the sparse count object. Cells are sampled from finite-coverage cells, weighted by `total_junction_reads`. Junctions are selected among high-coverage features using `n_cells_detected`, retaining junctions whose `event_id` belongs to a multi-junction LeafCutter/ATSE cluster. For each selected cell i and junction j , the numerator count is

$$k_{ij} = \text{cell_by_junction_matrix}_{ij},$$

and the denominator is

$$n_{ij} = \text{cell_by_cluster_matrix}_{ij}.$$

The denominator is the total count for the corresponding local splicing cluster, so k_{ij}/n_{ij} is the observed inclusion fraction for that junction whenever $n_{ij} > 0$.

3 Beta-binomial PSI estimates

For each junction j , a simple beta-binomial model is fit across cells:

$$k_{ij} \mid n_{ij}, p_{ij} \sim \text{Binomial}(n_{ij}, p_{ij}), \quad p_{ij} \sim \text{Beta}(\alpha_j, \beta_j).$$

The parameters (α_j, β_j) are estimated by maximum likelihood, parameterized by a mean and concentration. Given the fitted prior, the posterior parameters for an observed cell–junction pair are

$$a_{ij} = k_{ij} + \alpha_j, \quad b_{ij} = n_{ij} - k_{ij} + \beta_j.$$

The posterior PSI mean and standard error are

$$\hat{\psi}_{ij} = \frac{a_{ij}}{a_{ij} + b_{ij}}, \quad \hat{\text{se}}(\psi_{ij}) = \left\{ \frac{a_{ij}b_{ij}}{(a_{ij} + b_{ij})^2(a_{ij} + b_{ij} + 1)} \right\}^{1/2}.$$

The proximal Gaussian models are fit on the logit scale:

$$Y_{ij} = \text{logit}(\hat{\psi}_{ij}).$$

The standard error is transformed by the delta method,

$$S_{ij} = \frac{\hat{\text{se}}(\psi_{ij})}{\hat{\psi}_{ij}(1 - \hat{\psi}_{ij})}.$$

Entries with $n_{ij} = 0$ are treated as missing.

4 Proximal model comparison

Observed entries are split into train, validation, and test sets. The regularization parameter is selected by validation RMSE on held-out observed entries. Final performance is reported on an independent test split, both on the logit scale and after transforming predictions back to PSI.

Three proximal NND models are compared.

4.1 Scaling the proximal algorithm

The dense proximal implementation stores Y , X , FISTA auxiliary arrays, and SVD work arrays as dense $m \times n$ matrices. For the full splicing object, $m = 142315$ and $n = 89831$, so one float32 dense matrix is about 51 GB. Repeated dense SVDs at this scale are not viable.

The scalable implementation instead represents the iterate X in low-rank form,

$$X = U \text{diag}(d) V^\top,$$

and stores the observations as a sparse CSR matrix. For the homoskedastic observed-entry loss with unit step size, the proximal-gradient update is

$$X^+ = \text{SVT}_\lambda \{X - P_\Omega(X - Y)\} = \text{SVT}_\lambda \{X + P_\Omega(Y - X)\}.$$

The matrix inside the singular-value thresholding operator is therefore a low-rank matrix plus a sparse correction. The code exposes this object as a `scipy.sparse.linalg.LinearOperator`; matrix-vector products are

$$v \mapsto U \text{diag}(d)(V^\top v) + Cv, \quad u \mapsto V \text{diag}(d)(U^\top u) + C^\top u,$$

where $C = P_\Omega(Y - X)$ is sparse. A truncated sparse SVD is then computed with `svds`, and singular values below λ are thresholded away. Thus neither X nor the filled-in matrix is ever materialized densely.

The ARPACK SVD backend was then replaced by a randomized SVD backend over the same `LinearOperator`. Given a Gaussian test matrix Ω , the algorithm forms $Q = \text{qr}(A\Omega)$, applies one optional power iteration with $A^\top Q$ and $A(A^\top Q)$, and computes the small SVD of $Q^\top A$. On the full splicing matrix, this reduced a rank-20 proximal iteration from roughly 2–3 minutes with ARPACK to about 37–40 seconds for the homoskedastic runs after the first iteration.

The full sparse logit-PSI cache contains the CSR `indptr`, `indices`, logit-PSI values, and logit-SE values as memory-mapped arrays on the HDD. The cache is about 15 GB and covers all 1,280,029,347 observed entries.

4.2 Scalable Taylor scoring

Exact Taylor ELBO and LOO scoring would require row-wise covariance calculations in $n = 89831$ dimensions, which is not tractable. The full-matrix scorer therefore uses a diagonal Taylor approximation from the low-rank fitted factors. If $X = U \text{diag}(d)V^\top$, the prior curvature diagonal is approximated by

$$r_j = \lambda \sum_{\ell} \frac{V_{j\ell}^2}{\max(d_\ell, \epsilon)}.$$

For an observed entry with observation variance v_{ij} , the diagonal posterior precision approximation is

$$q_{ij} = v_{ij}^{-1} + r_j.$$

The diagonal Taylor ELBO surrogate uses the observed Gaussian data term, the diagonal log-determinant term $\frac{1}{2} \sum_{(i,j) \in \Omega} \log q_{ij}$, and the nuclear norm penalty. The LOO surrogate uses the local leverage

$$h_{ij} = v_{ij}^{-1} / q_{ij}$$

and reports the RMSE of $e_{ij}/(1 - h_{ij})$. These scores are scalable and use all observed entries, but they are diagonal approximations rather than the exact row-covariance Taylor scores.

An intermediate approximation keeps row-wise covariance structure in the fitted low-rank subspace and uses an isotropic completion off that subspace. Write the right singular vectors of the fitted matrix as V and the retained singular values as d_ℓ . The completed Gram approximation is

$$X^\top X \approx V \text{diag}(d_\ell^2) V^\top + \delta^2 (I - VV^\top),$$

with $\delta = \max_\ell d_\ell$ by default. The Taylor prior precision is then

$$B = \lambda V \text{diag}(d_\ell^{-1}) V^\top + \lambda \delta^{-1} (I - VV^\top) = \tau I + VCV^\top, \quad \tau = \lambda/\delta.$$

For row i , let D_i be the diagonal matrix of observed precisions in that row. The row covariance approximation is

$$\Sigma_i = (D_i + B)^{-1}.$$

This is evaluated by Woodbury in the retained rank r , using the sparse observed columns in each row. The cost is proportional to the number of observed entries in the scored rows times r^2 , so the current implementation uses random row subsampling for scoring. In the run below, 10000 rows were sampled, covering about 8.94×10^7 observed entries.

Homoskedastic model. This model ignores the beta-binomial standard errors in the X -update and estimates a single residual variance:

$$Y_{ij} \mid X_{ij}, \gamma^2 \sim N(X_{ij}, \gamma^2), \quad X \mid \gamma^2 \sim \text{NND}(\lambda/\sqrt{\gamma^2}).$$

The matrix X is updated by nuclear-norm proximal gradient, and γ^2 is updated using the Taylor/Hutchinson approximation described in the homoskedastic noise derivation.

Heteroskedastic model. This model uses the beta-binomial delta-method standard errors as known observation noise:

$$Y_{ij} \mid X_{ij}, S_{ij} \sim N(X_{ij}, S_{ij}^2).$$

It solves

$$\min_X \frac{1}{2} \sum_{(i,j) \in \Omega} \frac{(Y_{ij} - X_{ij})^2}{S_{ij}^2} + \lambda \|X\|_*.$$

Combined model. This model uses the beta-binomial standard errors but also estimates an additional homoskedastic residual variance g :

$$Y_{ij} \mid X_{ij}, S_{ij}, g \sim N(X_{ij}, S_{ij}^2 + g).$$

The X -update is the same efficient proximal update with effective standard errors $(S_{ij}^2 + g)^{1/2}$. The scalar g is updated by a Taylor diagonal-variance approximation, using Hutchinson probes for the trace terms when requested. The analysis grids over the effective nuclear-norm penalty rather than jointly estimating λ and g .

5 Debug runs

The main real-data debug run used a 600×200 block, 40 proximal iterations, and the lambda grid

$$\{0.01, 0.03, 0.1, 0.3, 1\}.$$

Sparse block extraction took about 11 seconds, beta-binomial PSI fitting about 1 second, and the proximal model comparison about 94 seconds.

model	selected λ	test logit RMSE	test PSI RMSE	noise estimate
homoskedastic	1.00	1.665	0.118	$\gamma^2 = 0.00441$
heteroskedastic	0.03	5.020	0.381	—
combined	1.00	4.692	0.297	$g = 0.00407$

On this block, the homoskedastic model performed best. The heteroskedastic and combined models were much worse, suggesting that the delta-method standard errors from the simple per-junction beta-binomial fit are not yet calibrated well enough on the logit scale, or that some high-precision entries are being overweighted relative to the low-rank signal.

5.1 All-cell, top-junction run

The literal full 142315×89831 proximal problem is not run as a dense matrix problem: it would require materializing many 12.8 billion-entry arrays and repeatedly computing SVDs. Instead, the full-cell run uses all 142315 cells and the top 20 high-coverage selected junctions. This keeps the cell axis complete while limiting the dense proximal block to 142315×20 . The same lambda grid $\{0.01, 0.03, 0.1, 0.3, 1\}$ and 40 proximal iterations were used. Sparse block extraction took about 109 seconds, beta-binomial PSI fitting about 10 seconds, proximal model comparison about 81 seconds, and the all-cell splicing-program cell-type classifiers about 152 seconds.

model	selected λ	test logit RMSE	test PSI RMSE	noise estimate
homoskedastic	1.00	2.319	0.083	$\gamma^2 = 5.40 \times 10^{-5}$
heteroskedastic	0.03	5.174	0.420	–
combined	1.00	3.846	0.173	$g = 0.450$

The all-cell top-junction result again favors the homoskedastic model. The combined model improves substantially over the heteroskedastic-only model but still does not match the homoskedastic fit on held-out PSI RMSE.

5.2 Literal full observed-matrix sparse proximal run

The sparse-plus-low-rank implementation was run on the literal full observed matrix, with shape 142315×89831 and 1,280,029,347 observed entries. The first cache construction streamed the whole HDF5 count object to fit per-junction beta moments and write logit-PSI/logit-SE memmaps. Subsequent proximal runs reuse this cache.

With rank cap 20 and $\lambda = 10000$, the first three full sparse SVT iterations took 127, 130, and 157 seconds, respectively. Continuing for three more iterations from the saved factors took 154, 127, and 152 seconds. The retained rank shrank from 6 after the first update to 2 after six updates, and the final observed logit RMSE was 2.145.

With rank cap 30 and $\lambda = 5000$, the first two full sparse SVT iterations took 226 and 192 seconds. The retained rank was 19 after the first update and 14 after the second update, with final observed logit RMSE 2.575. This lower penalty preserves more splicing structure but needs additional iterations for a fair comparison with the longer $\lambda = 10000$ run.

With the randomized SVD backend, a one-step lambda sweep over $\lambda \in \{5000, 10000\}$ was run for the homoskedastic, heteroskedastic, and combined noise models. Within each noise model, the diagonal Taylor ELBO selected $\lambda = 5000$. The diagonal LOO surrogate also selected $\lambda = 5000$ for the homoskedastic and combined models; for the heteroskedastic model the LOO scores for 5000 and 10000 were very close, with a slight preference for 10000 after one step.

Using the ELBO-selected $\lambda = 5000$, three randomized proximal iterations were run for each noise model. The corrected homoskedastic score estimates γ^2 from the final residuals for scoring, while treating λ as the effective proximal penalty during fitting.

model	rank	obs. logit RMSE	diag. LOO RMSE	diag. Taylor ELBO	noise
homoskedastic	10	2.335	231.3	-1.856×10^9	$\gamma^2 = 5.45$
heteroskedastic	18	4.596	320.5	-1.727×10^9	–
combined	18	4.548	368.6	-1.624×10^9	$g \approx 10^{-6}$

The diagonal Taylor ELBO favors the combined model, while the diagonal LOO surrogate and direct observed-entry reconstruction favor the homoskedastic model. Because the ELBO and LOO scores here use a diagonal approximation to the Taylor covariance, this disagreement should be interpreted as a diagnostic: the scalable scoring machinery is now in place, but the calibration of the heteroskedastic standard errors and diagonal curvature approximation still needs stress-testing.

The low-rank-plus-isotropic row covariance scorer was then evaluated on the same $\lambda = 5000$ fits using 10000 sampled rows.

model	obs. logit RMSE	row-iso LOO RMSE	row-iso ELBO	seconds
homoskedastic	2.335	15.293	1.767×10^{10}	10.6
heteroskedastic	4.595	5.958	-1.992×10^9	60.6
combined	4.547	7.216	1.066×10^9	98.1

This row-covariance LOO approximation is much less pathological than the purely diagonal LOO approximation. It favors the heteroskedastic model, despite its worse raw reconstruction RMSE, because it accounts for posterior uncertainty in the fitted low-rank subspace. The row-iso ELBO contains a large isotropic log-determinant contribution that differs strongly across noise models, so it should not yet be used as the sole selection rule without additional calibration or shared constants across model families.

5.3 All-cell, selected 10000-junction run

A larger all-cell subset was built by selecting 10000 junctions using both coverage and variability across medium-resolution cell types. For each junction, counts were aggregated by `medium_cell_type`, and the cell-type PSI variability was computed from the cell-type-specific $\sum k / \sum n$ estimates. Eligible junctions were required to belong to a multi-junction event and to have sufficient denominator counts in at least 8 cell types. Junctions were ranked by the sum of their percentile ranks for $\log(1 + n_{\text{cells detected}})$ and between-cell-type PSI variance.

The selected set contains 10000 junctions, spans 6497 splicing events and 4380 genes, and has median detection in 12338 cells. The resulting all-cell sparse cache has shape 142315×10000 with 263,931,532 observed entries.

A one-step lambda sweep over $\lambda \in \{1000, 3000, 5000, 10000\}$ was run for each noise model. The longer follow-up fits used $\lambda = 1000$ for the homoskedastic model, $\lambda = 3000$ for the heteroskedastic model, and $\lambda = 1000$ for the combined model. Each fit used 5 randomized proximal iterations with rank cap 50.

model	λ	rank	obs. logit RMSE	row-iso LOO RMSE	row-iso ELBO
homoskedastic	1000	13	1.534	26.749	1.901×10^9
heteroskedastic	3000	13	2.546	5.467	-3.104×10^8
combined	1000	2	2.115	9.779	1.500×10^9

The same pattern appears more cleanly on this subset: raw observed-entry RMSE favors the homoskedastic model, while the low-rank-plus-isotropic row LOO score favors the heteroskedastic model. The combined model is intermediate by LOO and raw RMSE. Thus, when the selected junctions are chosen to be both covered and cell-type-variable, the heteroskedastic uncertainty model looks more plausible under the row-covariance predictive criterion even though it does not minimize point-reconstruction error.

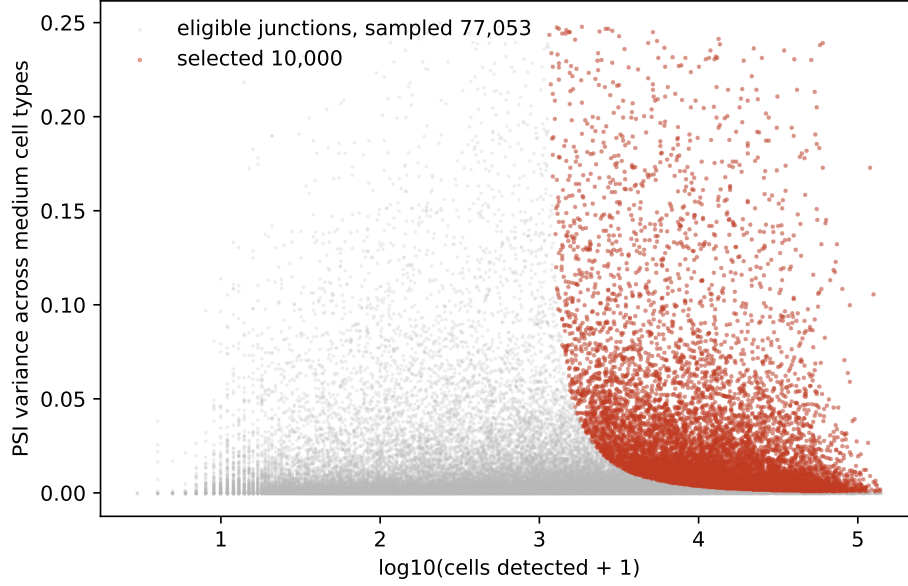


Figure 1: Selection of the 10000-junction subset. Grey points are eligible multi-junction features with sufficient medium-cell-type coverage; red points are the selected junctions. The selected subset emphasizes both coverage and between-cell-type PSI variability.

5.4 Logit-space versus PSI-space fitting

The sparse proximal runner was extended so that the same observed CSR pattern can be fit either on

$$Y_{ij} = \text{logit}(\hat{\psi}_{ij})$$

or directly on the bounded PSI scale $Y_{ij} = \hat{\psi}_{ij}$. The PSI-space standard errors were obtained from the existing logit-scale standard errors by the inverse delta-method relation

$$S_{ij}^{(\psi)} = S_{ij}^{(\text{logit})} \hat{\psi}_{ij} (1 - \hat{\psi}_{ij}).$$

For the PSI-space heteroskedastic model the regularization scale must be much larger, because the smallest allowed variance was 0.01^2 , giving a proximal step size of 10^{-4} . Thus a threshold of order 100 requires λ of order 10^6 .

The PSI-space all-cell, 10000-junction fits used $\lambda = 300$ for the homoskedastic and combined models and $\lambda = 10^6$ for the heteroskedastic model. Each fit used 5 randomized proximal iterations with rank cap 50. The homoskedastic PSI fit retained rank 7, the heteroskedastic fit retained rank 13, and the combined fit reached the rank cap after g shrank and the effective threshold dropped.

space	model	λ	rank	obs. RMSE	row-iso LOO RMSE
logit	homoskedastic	1000	13	1.534	26.749
logit	heteroskedastic	3000	13	2.546	5.467
logit	combined	1000	2	2.115	9.779
PSI	homoskedastic	300	7	0.276	27.599
PSI	heteroskedastic	10^6	13	0.539	3.198
PSI	combined	300	50	0.259	25.928

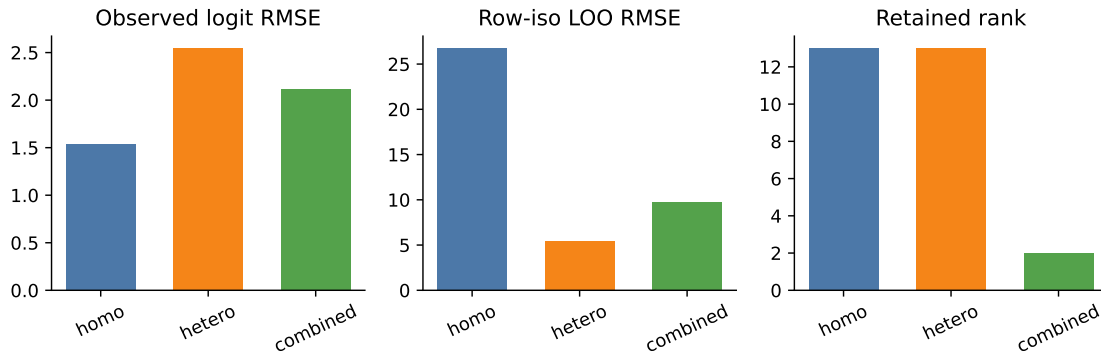


Figure 2: Model comparison on all cells and the selected 10000-junction subset. The homoskedastic model gives the best point reconstruction, while the heteroskedastic model gives the best low-rank-plus-isotropic row LOO score.

Observed-entry RMSE is not directly comparable across logit and PSI scales: the PSI residual is bounded while the logit residual is not. Within each fitted space, however, the qualitative pattern is stable. Homoskedastic or combined fits give better point reconstruction, while the heteroskedastic model gives the best low-rank-plus-isotropic row LOO score. The combined PSI fit should be interpreted cautiously because its retained rank is capped at 50, indicating that joint g and low-rank fitting is again reducing the effective regularization too aggressively.

The heteroskedastic PSI fit also exposed a practical step-size issue. With a hard SE floor of 0.01, the maximum precision is 10^4 , so the global proximal-gradient step is only 10^{-4} . This is mathematically consistent with the weighted Gaussian objective, but it makes the update nearly frozen by a small number of very high-precision observations. In the selected 10000-junction cache, the empirical PSI-SE quantiles are approximately 0.003 at 1%, 0.052 at 25%, and 0.101 at 50%. A follow-up run capped precision by raising the floor to the first-quartile PSI-SE, 0.052. With $\lambda = 100000$, this increased the step to 0.00272, retained rank 7 after five iterations, and gave observed PSI RMSE 0.375 with row-iso LOO RMSE 5.690. A median floor moved faster and gave lower point RMSE, but it caps half of the observed entries and is therefore a more aggressive change to the noise model. Thus the tiny heteroskedastic step is not an SVD or warm-start issue; it is a precision-capping issue. Quantile SE floors should be treated as part of the reported noise model.

To visualize whether the fitted low-rank splicing spaces preserve cell-type structure, UMAPs were computed from the atlas splicing programs and from fitted cell factor scores $U \text{diag}(d)$. The same 20000 sampled cells were embedded in each panel and colored by broad cell type. The UMAP and junction-selection plots are rendered as PDFs with rasterized point layers and subsampling, rather than pure vector graphics containing every point, to keep the document size manageable.

Cell-type separation was quantified directly in latent space using local k -nearest-neighbor purity and silhouette. For each embedding, 30000 cells were sampled and each cell's 15 nearest neighbors were queried in the standardized latent coordinates. The reported enrichment divides the observed same-label neighbor fraction by the same-label fraction expected from the label frequencies alone. Silhouette was computed on a 12000-cell subsample.

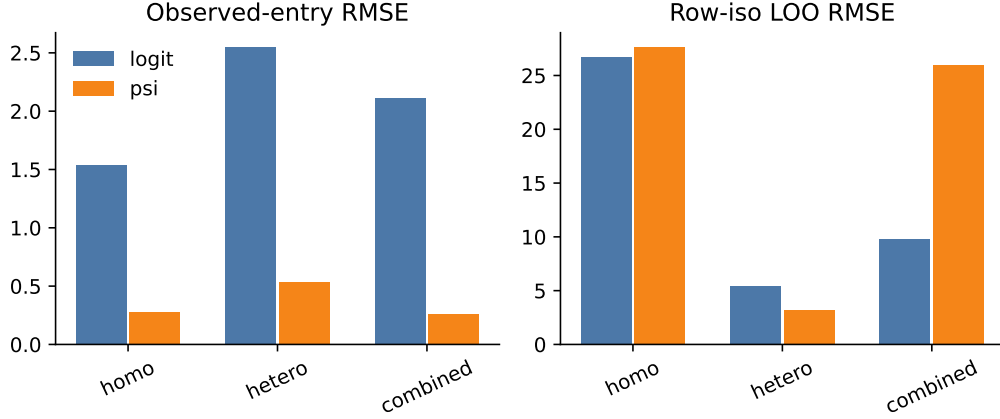


Figure 3: Comparison of logit-space and PSI-space proximal fits on all cells and the selected 10000-junction subset. RMSE values are only comparable within a value space; the row-iso LOO ordering favors the heteroskedastic uncertainty model in both spaces.

embedding	label	local purity	enrichment	silhouette
atlas SP	broad	0.728	3.57	-0.074
logit homo	broad	0.851	4.17	0.019
PSI homo	broad	0.818	4.01	0.021
PSI hetero q25	broad	0.815	4.00	0.006
atlas SP	medium	0.563	6.62	-0.097
logit homo	medium	0.696	8.18	-0.022
PSI homo	medium	0.636	7.48	-0.059
PSI hetero q25	medium	0.627	7.37	-0.092

The fitted proximal factors therefore show strong local cell-type structure: broad-cell-type neighbor purity is about 0.81–0.85 for the fitted spaces against a random-label baseline of 0.20, and medium-cell-type purity is about 0.62–0.70 against a baseline of 0.085. Silhouette is small and becomes negative for medium and specific labels, so the separation should be understood as local manifold organization rather than clean globally separated clusters.

To interpret which factors drive broad cell-type organization, factor usage was averaged within each broad cell type. Cell-level factor scores were first standardized within each embedding. Because singular vectors are sign ambiguous, each fitted factor was oriented so that the cell type with the largest absolute mean usage is positive before plotting.

5.5 Nonnegative factors from denoised proximal fits

For interpretability, NMF and semi-NMF were fit as post-processing layers on the PSI-scale proximal estimates. If

$$X_{\text{prox}} = U \text{diag}(d) V^{\top}$$

is a fitted proximal estimate, the factorization target was generated in row batches as

$$\tilde{X}_i = \min\{1, \max\{0, X_{\text{prox},i}\}\}.$$

Mini-batch NMF fit

$$\tilde{X} \approx WH, \quad W \geq 0, \quad H \geq 0,$$

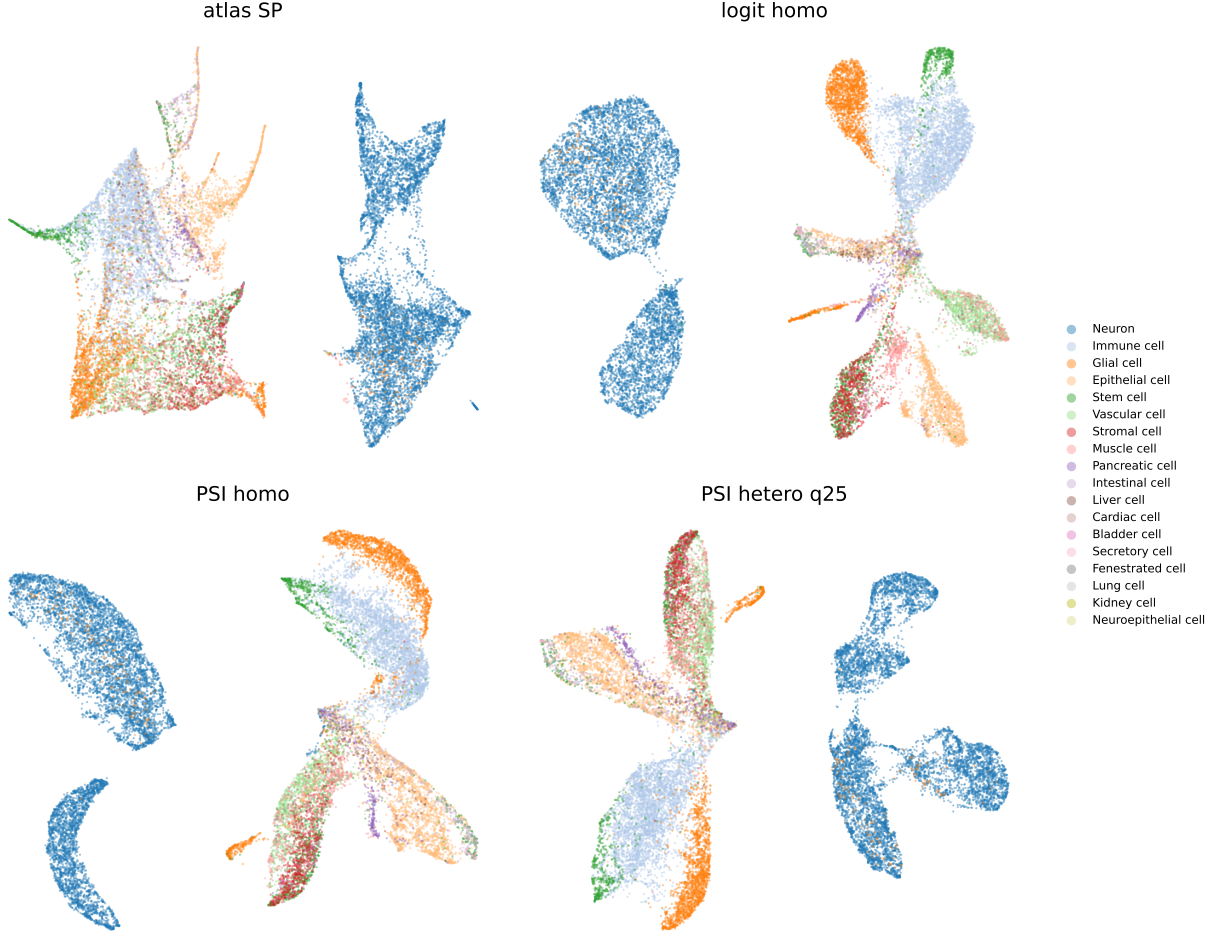


Figure 4: UMAPs of splicing latent spaces colored by broad cell type. The panels compare the published atlas splicing programs, logit-space homoskedastic factors, PSI-space homoskedastic factors, and PSI-space heteroskedastic factors with a first-quartile SE floor. All panels use the same sampled cells.

with 20 nonnegative programs. A semi-NMF variant fit

$$\tilde{X} \approx WH, \quad W \geq 0, \quad H \in \mathbb{R}^{20 \times p},$$

so cell usages are nonnegative while junction loadings can be signed. The semi-NMF update alternated between projected-gradient NNLS updates for row batches of W and a signed ridge least-squares update for H . Both algorithms never materialized the full 142315×10000 denoised matrix: each update generated a dense cell-by-10000-junction block from the low-rank factors, clipped it to $[0, 1]$, and then discarded the block. After fitting, cell usages W were computed for all cells in row batches.

target	method	rank	train sec.	transform sec.	eval RMSE
PSI homo	NMF	20	40.5	26.4	0.014
PSI hetero q25	NMF	20	40.7	27.6	0.009
PSI homo	semi-NMF	20	21.3	5.2	0.0017
PSI hetero q25	semi-NMF	20	21.3	5.1	0.0014

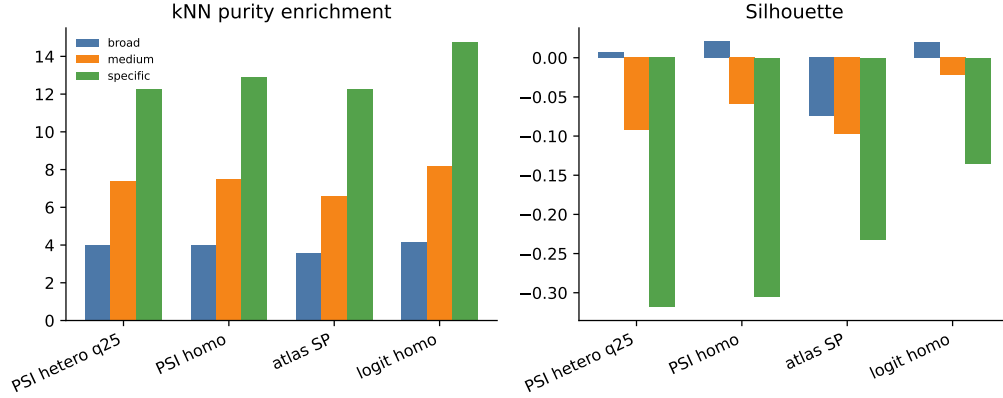


Figure 5: Cell-type separation metrics in splicing latent spaces. Local neighbor purity shows strong enrichment for broad, medium, and specific labels; silhouette is much weaker, indicating overlapping manifolds rather than well-separated convex clusters.

Both approximations therefore scale comfortably to all cells and the selected 10000 junctions. Semi-NMF reconstructs the denoised proximal target more closely because signed junction loadings are less restrictive. Medium-cell-type local purity in usage space was 0.514–0.549 for strict NMF and 0.549–0.576 for semi-NMF, compared with a random-label baseline of 0.085. Thus the nonnegative cell usages preserve meaningful cell-type neighborhood structure, although less strongly than the signed proximal factors.

embedding	medium purity	enrichment	silhouette
NMF PSI homo	0.514	6.04	-0.195
NMF PSI hetero q25	0.549	6.46	-0.211
Semi-NMF PSI homo	0.549	6.45	-0.213
Semi-NMF PSI hetero q25	0.576	6.77	-0.203

6 Cell-type analysis from splicing programs

To assess whether splicing defines cell types, a multinomial logistic classifier was trained to predict cell-type labels from the 20 fitted splicing program coordinates in the atlas. The final run used all available cells with a held-out 25% test split. Class weights were balanced in the logistic regression.

label	classes	accuracy	balanced accuracy	majority accuracy
broad cell type	11	0.767	0.686	0.255
medium cell type	58	0.514	0.348	0.217
specific cell type	132	0.382	0.263	0.085

The 20 splicing programs are therefore strongly informative for broad cell identity, reasonably informative for medium-resolution cell types, and still substantially above the majority-class baseline for specific cell types.

Using all cells, the corresponding metrics were: broad cell type accuracy 0.771 and balanced accuracy 0.698; medium cell type accuracy 0.524 and balanced accuracy 0.379; specific cell type accuracy 0.381 and balanced accuracy 0.281.

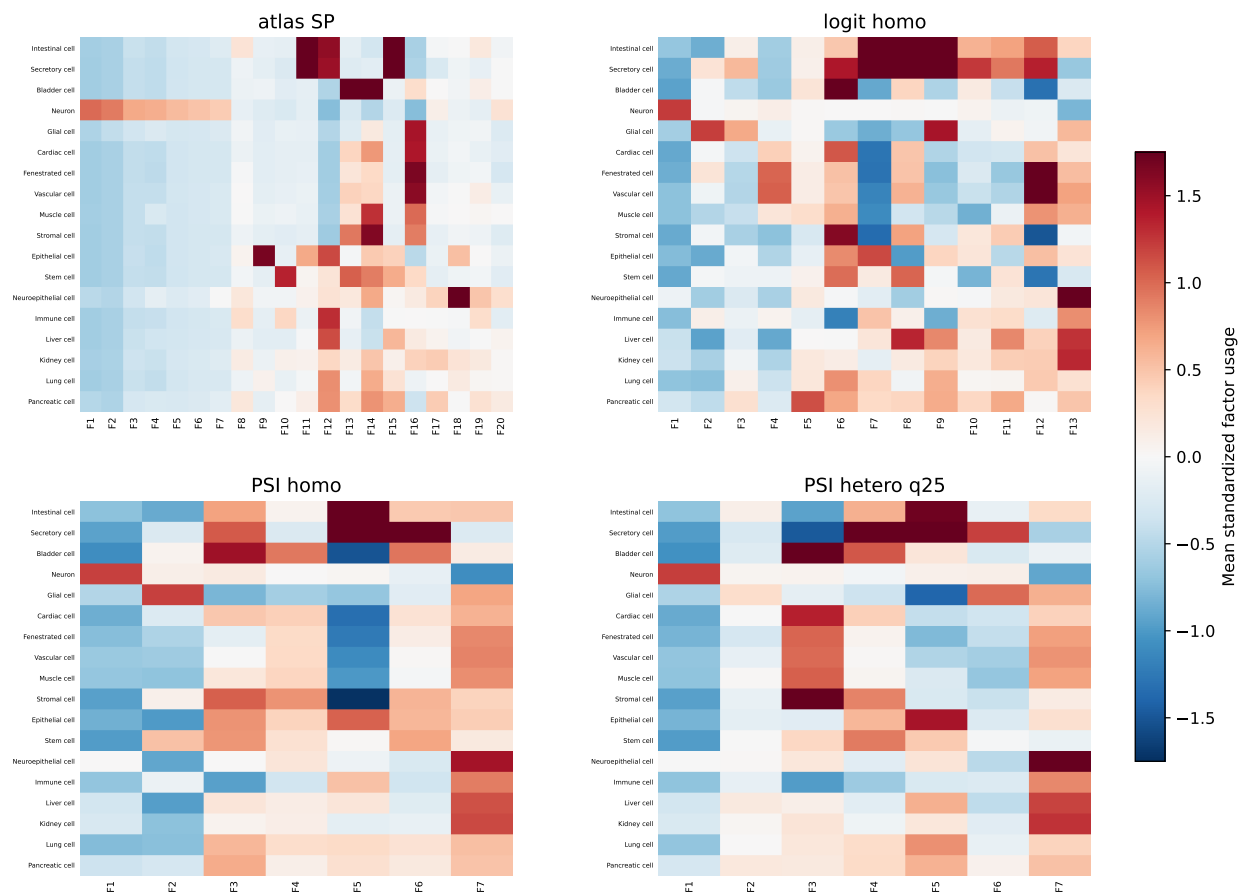


Figure 6: Mean standardized factor usage by broad cell type. Rows are broad cell types ordered by clustering the atlas splicing-program means. The panels compare atlas splicing programs, logit-space homoskedastic factors, PSI-space homoskedastic factors, and PSI-space heteroskedastic factors with a first-quartile SE floor.

7 Reproducibility

The analysis code is in `analyses/splicing_proximal/`. The command-line debug run was:

```
SPLICING_MAX_CELLS=600 \
SPLICING_MAX_JUNCTIONS=200 \
SPLICING_PROX_ITER=40 \
SPLICING_LAMBDA_GRID=0.01,0.03,0.1,0.3,1 \
SPLICING_CELLTYPE_MAX_CELLS=60000 \
~/venvs/jax/bin/python analyses/splicing_proximal/run_full_analysis.py
```

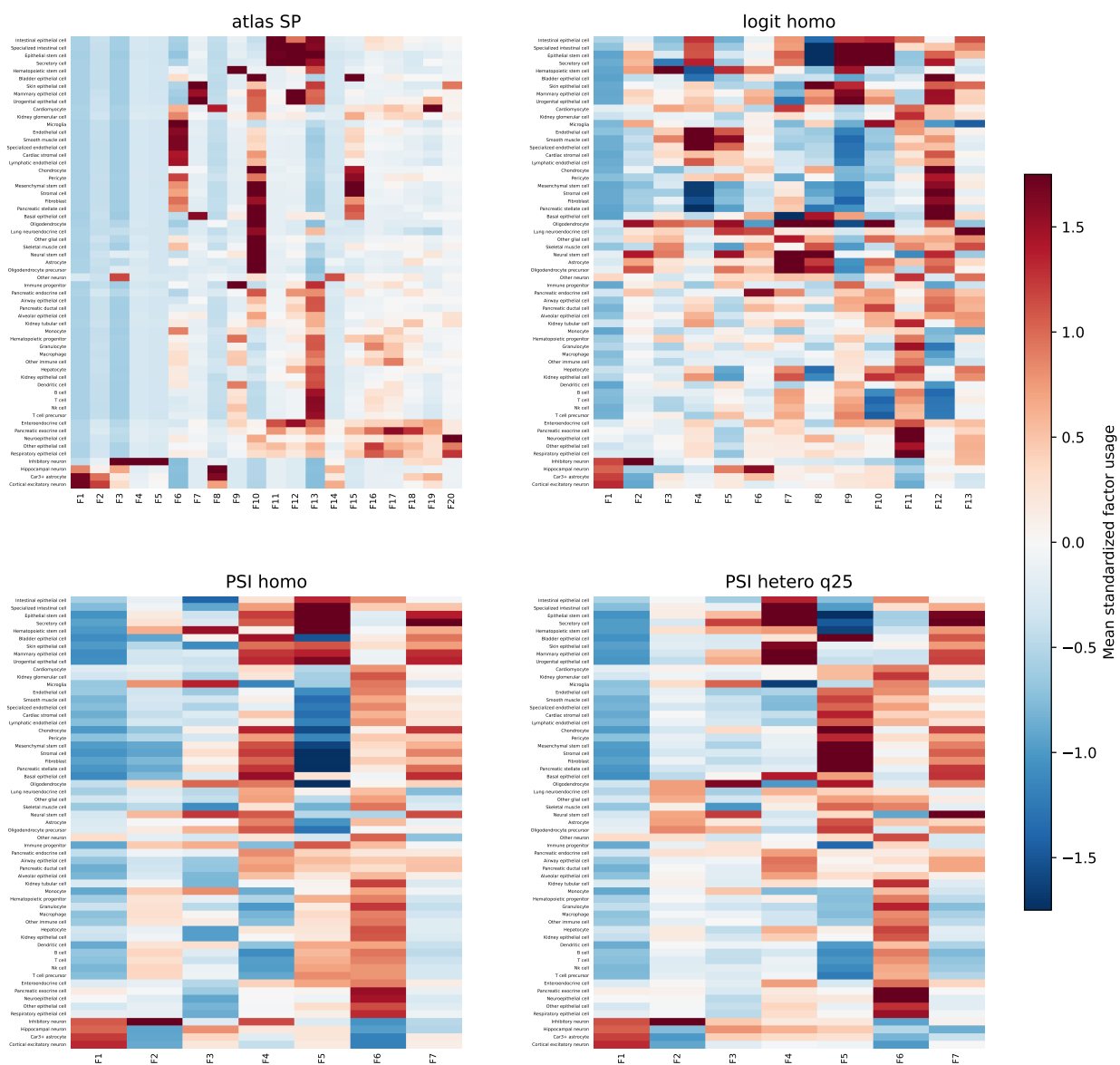


Figure 7: Mean standardized factor usage by medium cell type. The same factor orientation and row-clustering convention as Figure 6 is used.

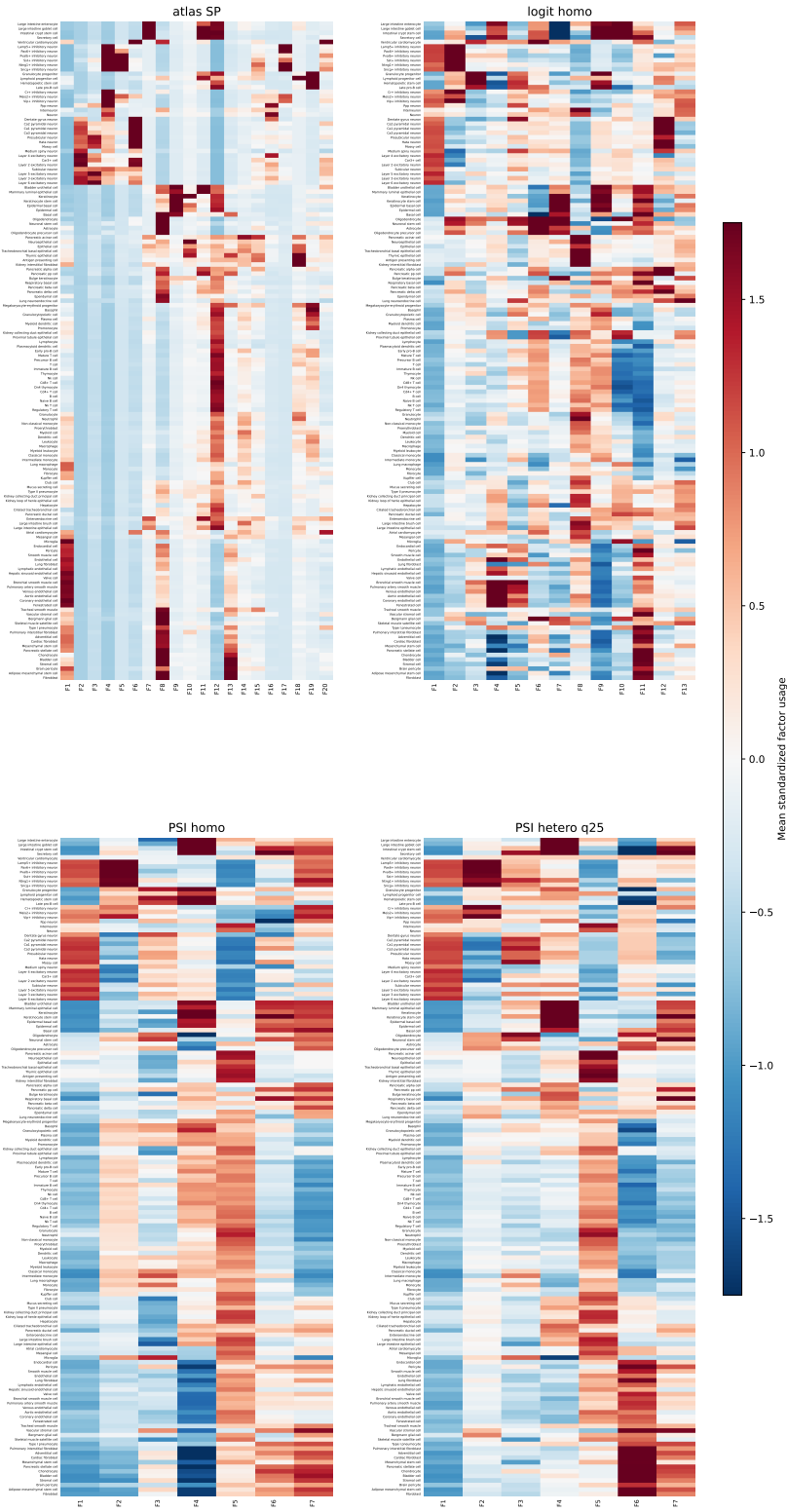


Figure 8: Mean standardized factor usage by specific cell type. This higher resolution view contains 145 cell-type labels and is intended for zoomed inspection in the PDF.

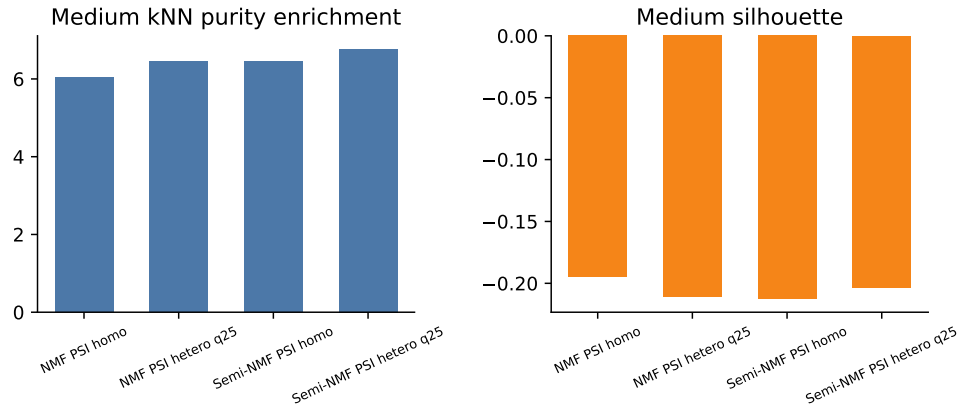


Figure 9: Medium-cell-type separation metrics for nonnegative usage coordinates. The NMF and semi-NMF usage spaces retain substantial local cell-type enrichment, with weaker global silhouette scores.

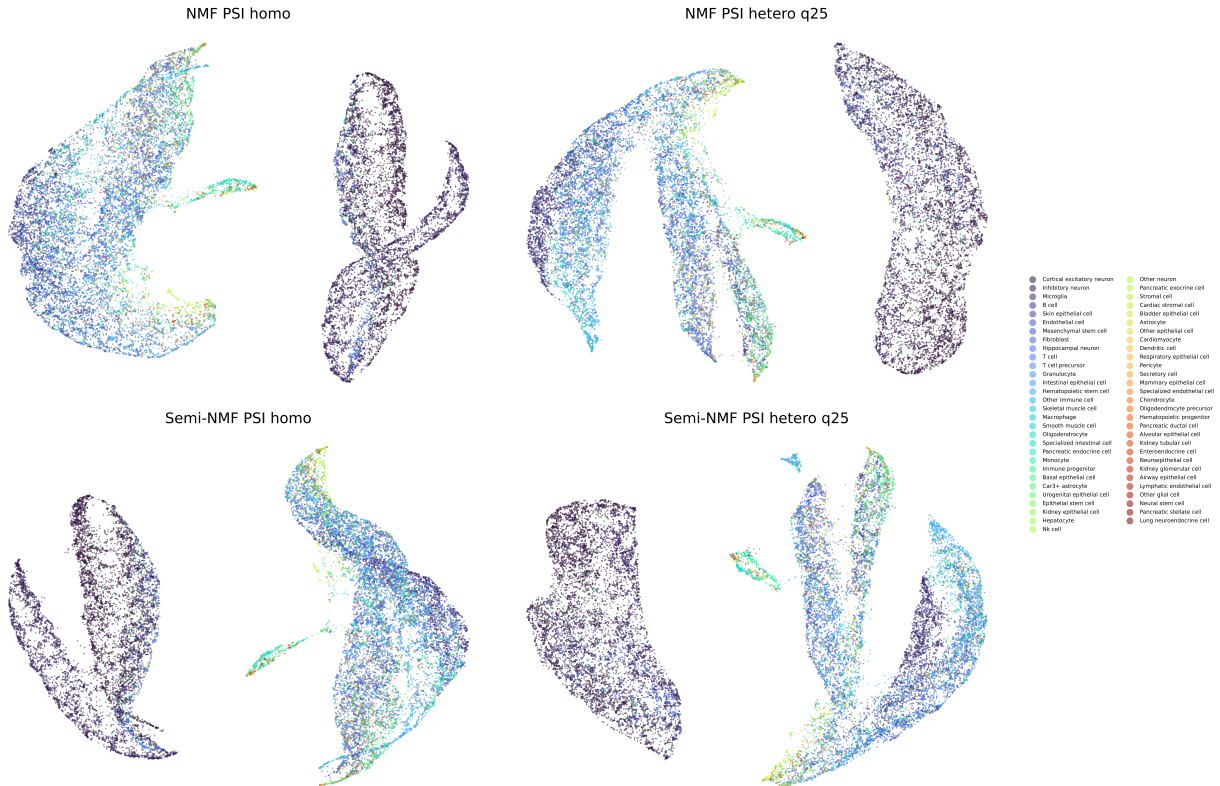


Figure 10: UMAPs of rank-20 nonnegative cell usage coordinates colored by medium cell type. Panels compare strict NMF and semi-NMF post-processing of the PSI-homoskedastic and PSI-heteroskedastic q25 proximal targets.

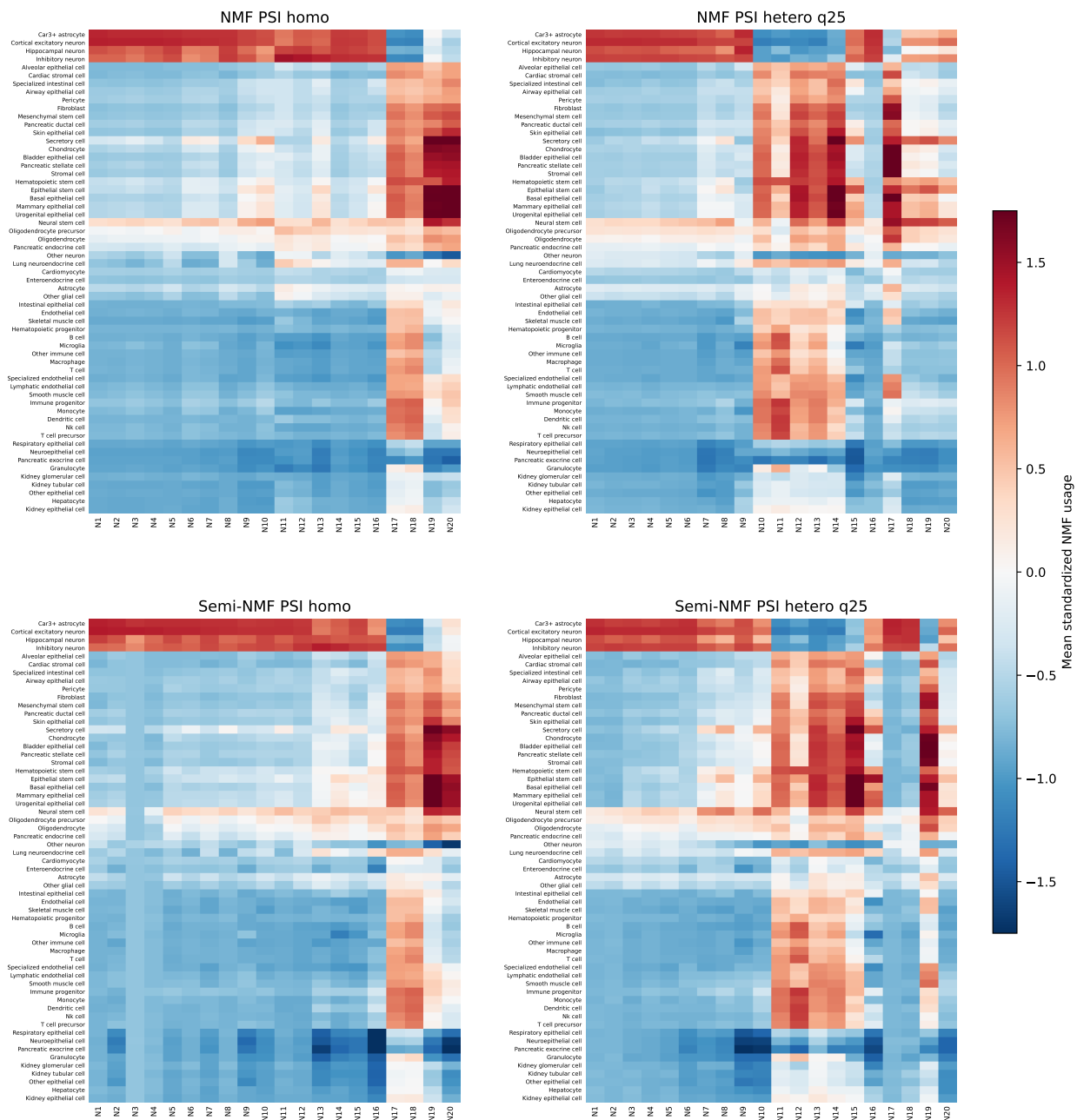


Figure 11: Mean standardized rank-20 nonnegative usage by medium cell type. Rows are medium cell types clustered by the PSI-homoskedastic NMF means.

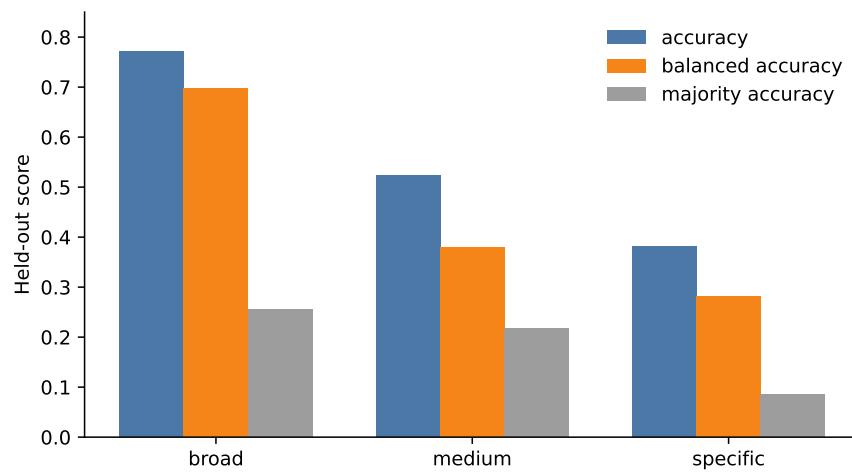


Figure 12: Cell-type prediction from the 20 fitted mouse splicing programs. Splicing programs are strongly predictive of broad cell type and remain informative for medium and specific cell-type labels relative to the majority-class baseline.